

Ashfak Hossain Evan

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EDUCATION

American International University Bangladesh Expected December 2026
BSc in Computer Science and Engineering

PROJECTS

shortn — Distributed URL Shortener

Go, PostgreSQL, Redis, Kubernetes, Terraform

github.com/Ashfak-Hossain/shortn • Live: shortn.ashfak.dev

- Built a production-grade URL shortener with a Redis cache-aside read path, PostgreSQL-backed storage, and a coordination-free Snowflake ID generator (sqids-obfuscated) for collision-free horizontal scaling across stateless replicas.
- Designed an asynchronous click-analytics pipeline on Redpanda (Kafka API) with exactly-once delivery via same-transaction offset commits, decoupling analytics writes from the redirect hot path.
- Found and fixed a connection-queueing bottleneck via k6 load testing — a missing nginx upstream keepalive — cutting redirect **p99 latency from 172ms to 11ms** with median latency unchanged.
- Hardened the system for failure: circuit breaker on Postgres, distributed rate limiting, idempotency keys, and fail-open behavior on cache/broker outages.
- Deployed to Kubernetes via Helm + ArgoCD (GitOps) with HPA autoscaling and zero-downtime rollouts; infrastructure declared in Terraform; full observability with Prometheus, Grafana, Loki, and Tempo via OpenTelemetry.

odyssey — 2D Platformer Engine

C++, OpenGL, GLUT

github.com/Ashfak-Hossain/odyssey

- Built a 2D platformer from raw OpenGL and GLUT calls with no external game or rendering library, implementing the render/input/update loop directly on GLUT's callback model.
- Organized the engine into a class-based C++ architecture, with class diagrams auto-generated via clang-uml and PlantUML to keep design and code in sync.

TECHNICAL SKILLS

Languages: C++, Go, TypeScript, JavaScript, C#

Frameworks & Libraries: React, Next.js, Express, NestJS

Data & Infrastructure: PostgreSQL, Redis, Docker, Kubernetes, Terraform

Tools & Practices: Git, REST API Design, CI/CD (GitHub Actions), Data Structures & Algorithms

ADDITIONAL

Competitive Programming: [Codeforces](https://codeforces.com/) (_Berlin_) • [LeetCode](https://leetcode.com/) (Berlin11) • [CodeChef](https://www.codechef.com/) (evan42)

Algorithms Notebook: public digital garden on math, algorithms, and competitive programming, built with Quartz/TypeScript and deployed via GitHub Pages — ashfak-hossain.github.io/codex